gridlocks – can you unlock the grid?

Compounds ending in ‘ate’

If a compound has 3 elements in it with oxygen as one of them the names often end in ‘ate’ (if hydrogen is with the oxygen it can be a hydroxide rather than an ‘ate’). Before you answer the puzzles below fill in the table of compounds using:

silver nitrate **copper sulfate** **zinc carbonate** **sodium chlorate**

Name	Elements in the compound
calcium carbonate	calcium, carbon and oxygen
	copper, sulfur and oxygen
	silver, nitrogen and oxygen
sodium phosphate	sodium, phosphorus and oxygen
potassium manganate	potassium, manganese and oxygen
	sodium, chlorine and oxygen
potassium chromate	potassium, chromium and oxygen
	zinc, carbon and oxygen

Gridlock 1

Each row, column and 2 x 2 box contains information about the first four compounds listed above. Use your problem solving skills and the answers in the table above to fill in the blank boxes.

compound		made up of the elements	
calcium carbonate			sodium, phosphorus and oxygen
	sodium phosphate		
			calcium carbonate
		copper sulfate	

made up of the elements	compound

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Gridlock 2

Each row, column and 2 x 2 box contains information about each of the last four compounds in the table.

compound		made up of the elements	
sodium chlorate			
		zinc, carbon and oxygen	
		potassium manganate	
	potassium, chromium and oxygen		

made up of the elements	compound

Gridlock 3

You will need to identify which four compounds are in this puzzle.

compound		made up of the elements	
	zinc carbonate		potassium, manganese and oxygen
			zinc, carbon and oxygen
copper, sulfur and oxygen			silver nitrate

made up of the elements	compound